

MA540 Project 4

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Problem 1

Statement

We consider the height model $Y_i = \theta_0 + \theta_1 t_i + \theta_2 t_i^2 + \varepsilon_i$. We will use the data compiled in Table 1, the design matrix X given in class, the estimates $\boldsymbol{\theta} = [143.4048, 1.7389, -2.4892 \times 10^{-2}]^T$, $\sigma^2 = 0.2687$, and

$$\mathbf{V} = \begin{bmatrix} 4.3500 \times 10^{-1} & -1.8232 \times 10^{-1} & 1.5993 \times 10^{-2} \\ -1.8232 \times 10^{-1} & 9.1710 \times 10^{-2} & -8.7232 \times 10^{-3} \\ 1.5993 \times 10^{-2} & -8.7232 \times 10^{-3} & 8.7232 \times 10^{-4} \end{bmatrix}$$

to compute and plot 2σ and 95% frequentist prediction intervals. We will then compute these intervals over the extrapolatory regime of 10 to 15.

Solution

After computing our model, we compute the frequentist prediction intervals by

$$\left[\hat{Y}_{\mathbf{x}_*} \pm t_{n-p, 1-\alpha/2} \cdot \sigma \sqrt{1 + \mathbf{x}_* \mathbf{V} \mathbf{x}_*^T} \right]$$

using $\alpha = 0.05$, $n = 9$, $p = 3$, and our initial estimates. We also compute the constant 2σ intervals using the estimate. We then compute these over $t_i = 1, \dots, 15$. Figure 1 shows the 2σ and prediction intervals graphed alongside the data. Since they are based on a constant value, the 2σ values stay the same width in the extrapolatory regime. While they seem to hold 95% of the original data, we do not expect that the 2σ intervals have at least probability 95% of containing a future observable $Y_{x_i^*}$ in the extrapolatory regime, as the uncertainty compounds. By contrast, the prediction intervals widen considerably, taking into account the propagation of the uncertainty. Thus, we do expect that these intervals have at least a 95% probability of containing a future observable $Y_{x_i^*}$ in the extrapolatory regime, if measured.

Problem 2

Statement

We consider the steady state heat model

$$\begin{aligned} \frac{d^2 u_s}{dx^2} &= \frac{2(a+b)}{ab} \frac{h}{k} [u_s(x) - u_{amb}] \\ \frac{du_s}{dx}(0) &= \frac{\Phi}{k} \quad , \quad \frac{du_s}{dx}(L) = \frac{h}{k} [u_{amb} - u_s(L)] \end{aligned}$$

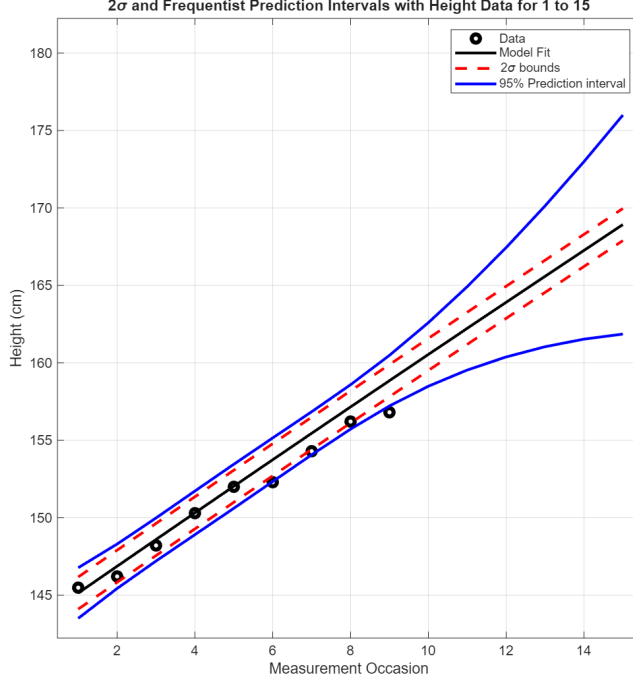


Figure 1: 2σ and Frequentist Prediction Intervals for Height Model

with a subset of the data from Table 3.2:

x (cm)	22	26	30	34	38	42	46	50	54
Temp ($^{\circ}\text{C}$)	57.96	50.90	44.84	39.75	36.16	33.31	31.15	29.28	27.88

We will employ DRAM to construct densities for Φ , h , and ε based on this subset of the data and the aluminum thermal conductivity value $k = 2.37 \frac{\text{W}}{\text{cm} \cdot ^{\circ}\text{C}}$. We also take $u_{amb} = 22.28$ from a previous project. We will then sample from these densities to construct credible and prediction intervals for the full domain of $[10, 66]$, using `mcmcpruned` and `mcmcprunedplot`. We will then graph these intervals with the complete data set from Table 3.2.

Solution

We use OLS to construct parameter estimates and a covariance matrix $\mathbf{V}$ to initialize DRAM. We then run `mcmcrun` on our model for $N = 10000$ iterations. Figure 2 shows the chains for each of the parameters under DRAM. Note that the chain for ε represents sampled σ^2 values for each iteration, resulting in strictly positive values. We note that each chain exhibits the "fuzzy caterpillar" appearance, indicating minimal stagnation and high exploration.

Then, we use `mcmcpruned` to sample from these densities for $N = 1000$ iterations in order to construct credible and prediction intervals for $x \in [10, 66]$. Figure 3 shows the credible and prediction intervals graphed with the complete dataset from Table 3.2. We note that the intervals are very tight, so a zoomed portion is shown. The prediction intervals contain all the extrapolatory points that weren't in the sampling subset, which follows from the uncertainty propagation theory for these Bayesian intervals. The credible intervals give us an idea of the uncertainty of our model mean.

Figure 2: Chains Under DRAM

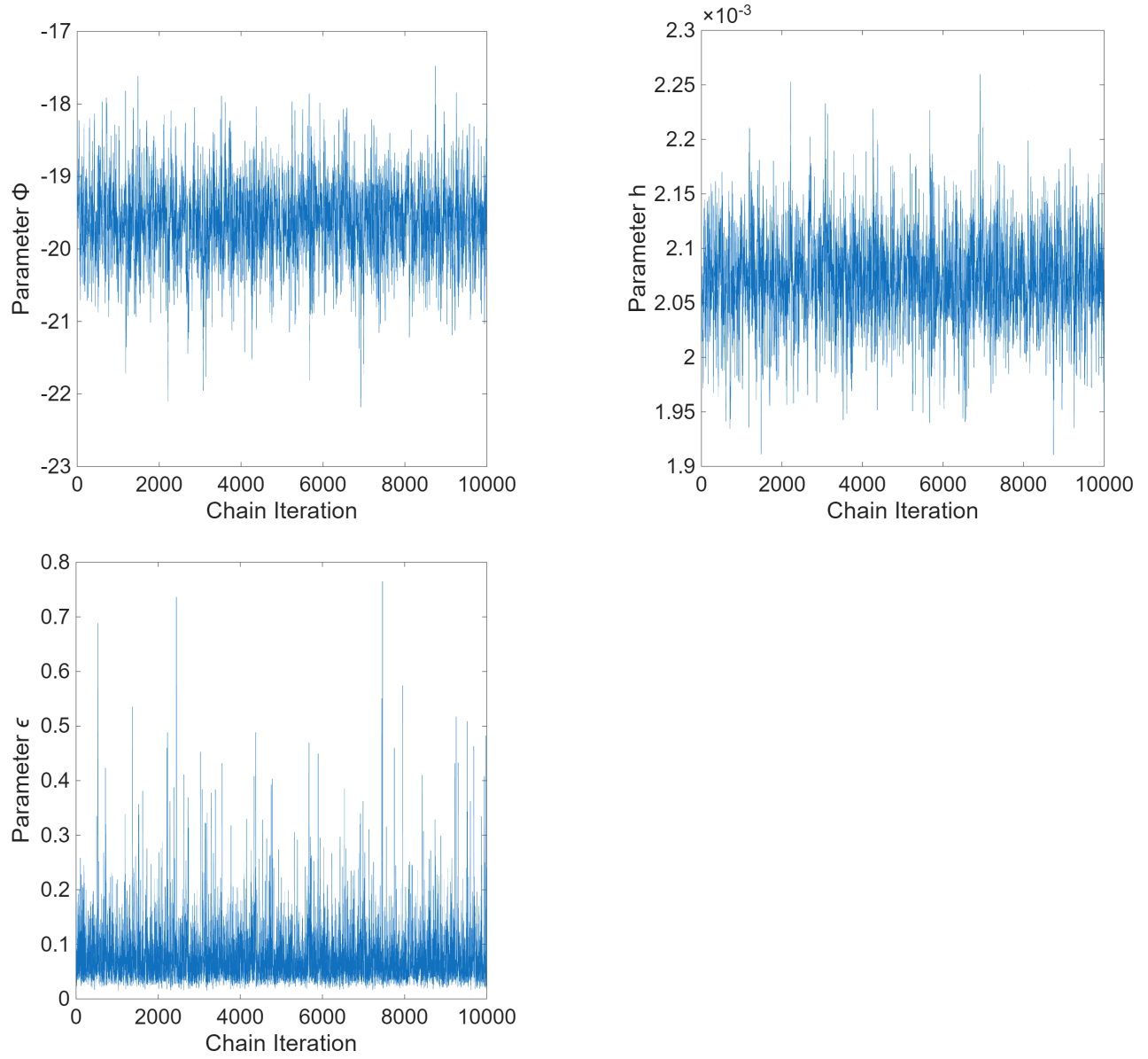
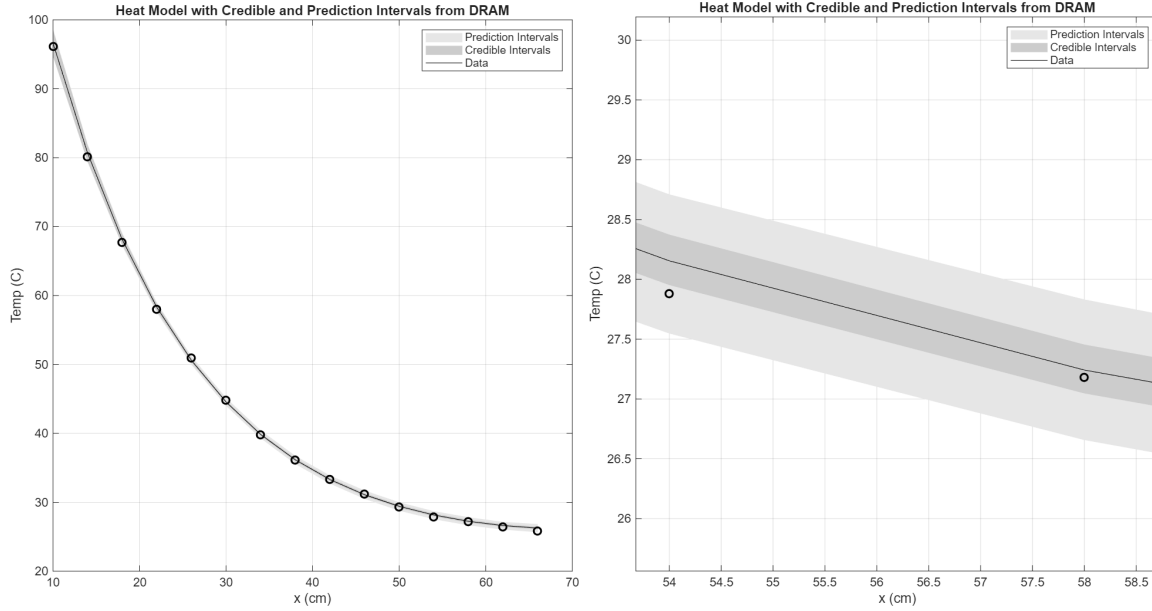


Figure 3: Bayesian Credible and Prediction Intervals for Heat Model



Problem 3

Statement

We now consider the SIR model

$$\begin{aligned}\frac{dS}{dt} &= \mu(N - S) - \eta kIS \quad , \\ \frac{dI}{dt} &= \eta kIS - (\gamma + \mu)I \quad , \\ \frac{dR}{dt} &= \gamma I - \mu R \quad ,\end{aligned}$$

with parameters $\beta, \gamma, \mu \in [0, 1]$. We will employ DRAM to compute parameter chains, which will then be used to construct 95% credible and prediction intervals for $I(t)$ on two different data sets using `mcmcpruned`. The first (a) will be the data from `SIR_integer.txt`, with initials $S(0) = 900, I(0) = 100, R(0) = 0$. The second (b) will be the data from Table 3, with initials $S(0) = 730, I(0) = 3, R(0) = 0$ and $\mu = 0$. We will graph the intervals for each with the corresponding data.

Solution

We begin with the first data set (a). We use OLS to construct parameter estimates and complex step to construct a covariance matrix $\mathbf{V}$ to initialize DRAM. We then run `mcmcrun` on this model for $N = 10000$ iterations. Figure 4 shows the chains for each parameter under DRAM. We note that each chain exhibits the "fuzzy caterpillar" appearance, indicating minimal stagnation and high exploration.

We then use `mcmcpruned` to sample from these densities for $N = 1000$ iterations in order to construct credible and prediction intervals over the data. Figure 5 shows the credible and prediction intervals over the original data from `SIR_integer.txt`. Again, the intervals are very tight, so a zoomed portion

Figure 4: Chains Under DRAM

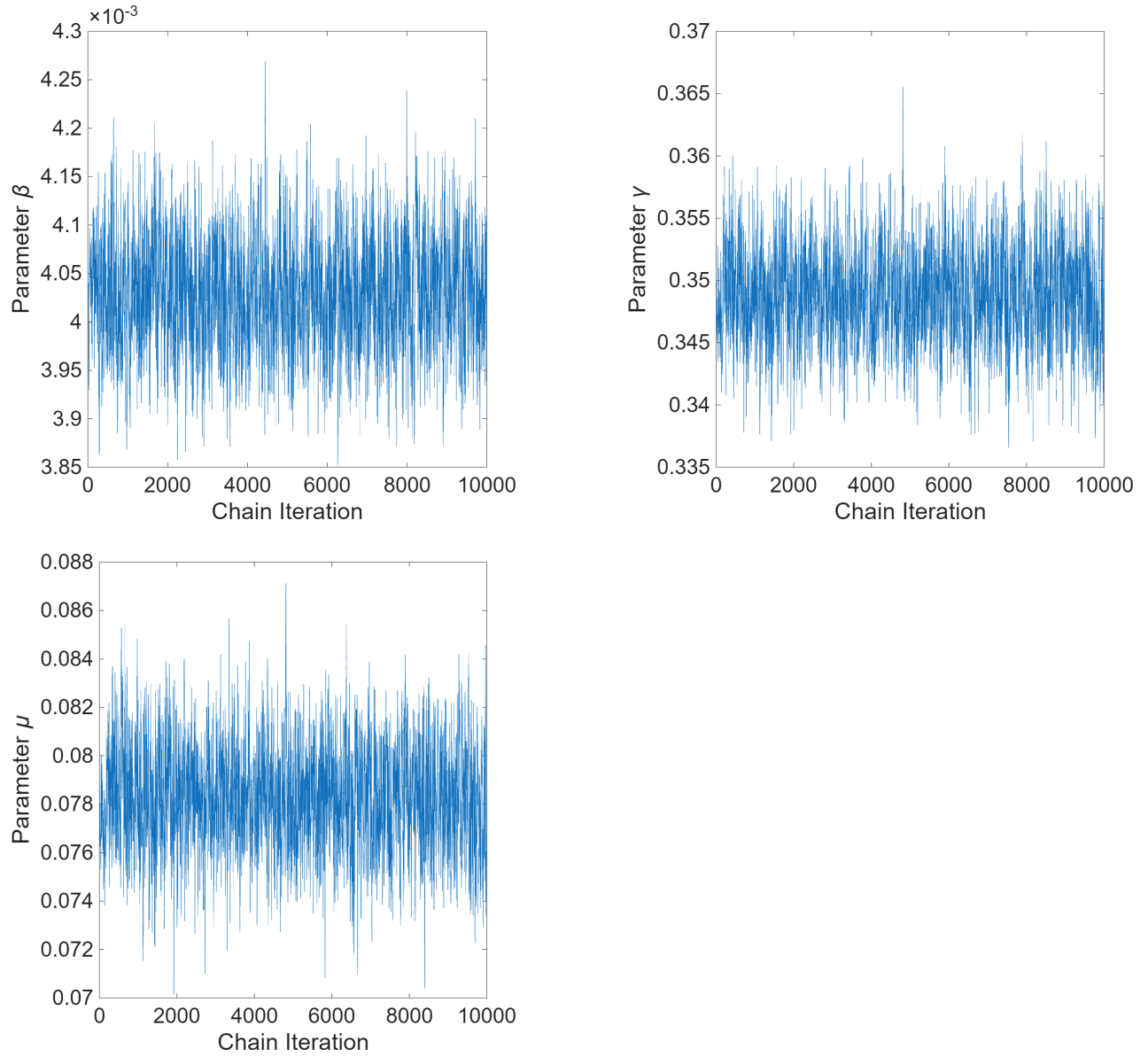
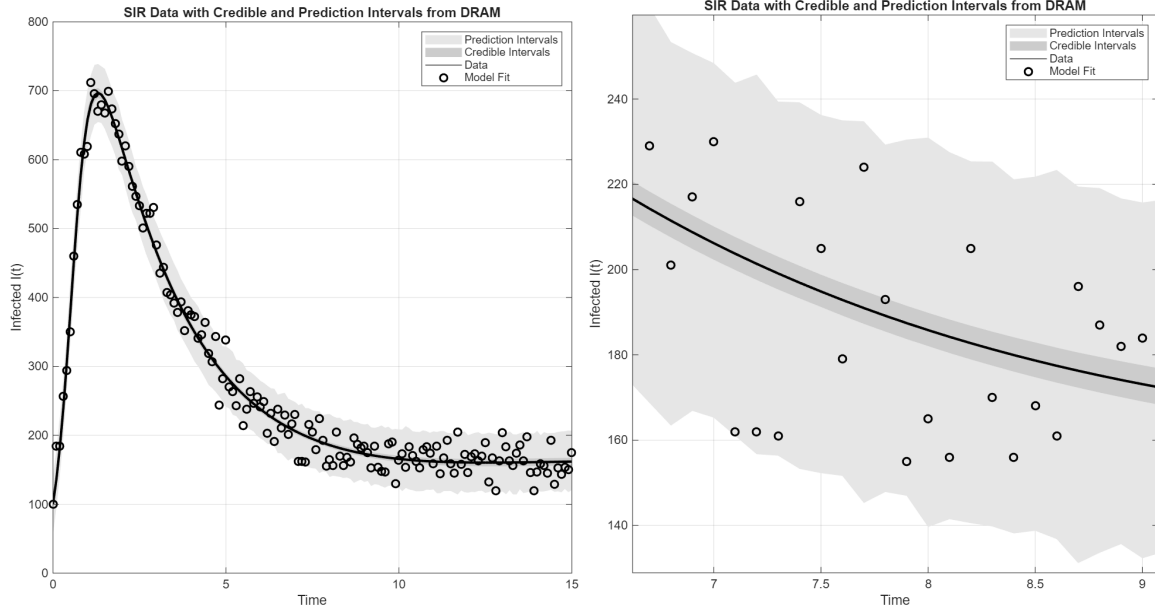


Figure 5: Bayesian Credible and Prediction Intervals for SIR Model from `SIR_integer.txt`



is shown. We note that 95% of the original data appears to be contained in the prediction intervals, which we expect, as a future observable datapoint has at least a 95% probability of being contained. The credible intervals are very tight around the model, indicating low expected uncertainty in our model itself based on the uncertainty of the parameters.

Now, we consider the second data set (b). We again use OLS to initialize parameters for DRAM, but do not specify a covariance matrix. Figure 6 shows the chains for β, γ under DRAM. Again, we see minimal stagnation and high exploration for both.

We then sample from these densities and construct intervals in the same process as (a), using 200 equally spaced points t_i within $(0, 13)$. Figure 7 shows the credible and prediction intervals over the original influenza data. With a sparse dataset, we note wider prediction intervals, as there is greater uncertainty in regards to future observable data points. Similarly, our credible intervals also reflect increased uncertainty in our model mean.

Problem 4

Statement

We consider the same SIR model from Problem 3(a), with data from `SIR_integer.txt` and initials $S(0) = 900, I(0) = 100, R(0) = 0$. We will use complex-step approximation to estimate the sensitivity matrix $\mathbf{S}$ for the nominal parameter values $\theta^* = [0.0040, 0.3488, 0.0782]^T$, using the estimate $\sigma^2 = 465.67$. We will then find the covariance matrix $\mathbf{V}$ to construct response and observation matrices by $\text{var}[\mathbf{f}(\theta)] = \mathbf{S}\mathbf{V}\mathbf{S}^T$, $\text{var}[\mathbf{Y}] = \mathbf{S}\mathbf{V}\mathbf{S}^T + \sigma^2\mathbf{I}_{n \times n}$. We will then compute the square roots of the diagonal elements of each matrix to construct 2σ intervals for the model response and the observed data, respectively. We will graph these with the model and observed data, and compare with the credible and prediction intervals computed in Problem 3.

Figure 6: Chains under DRAM

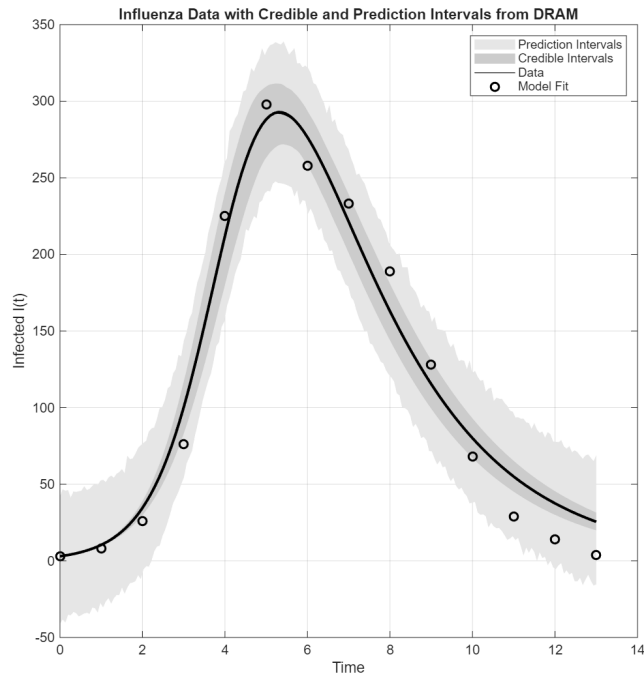
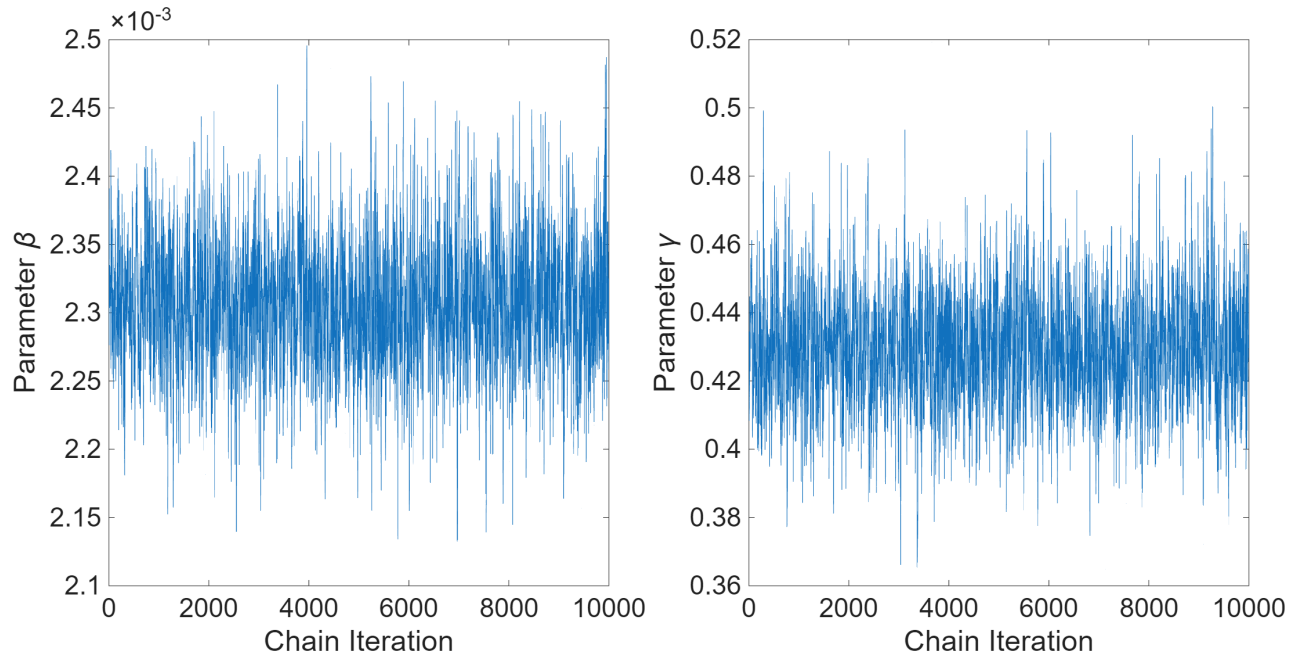
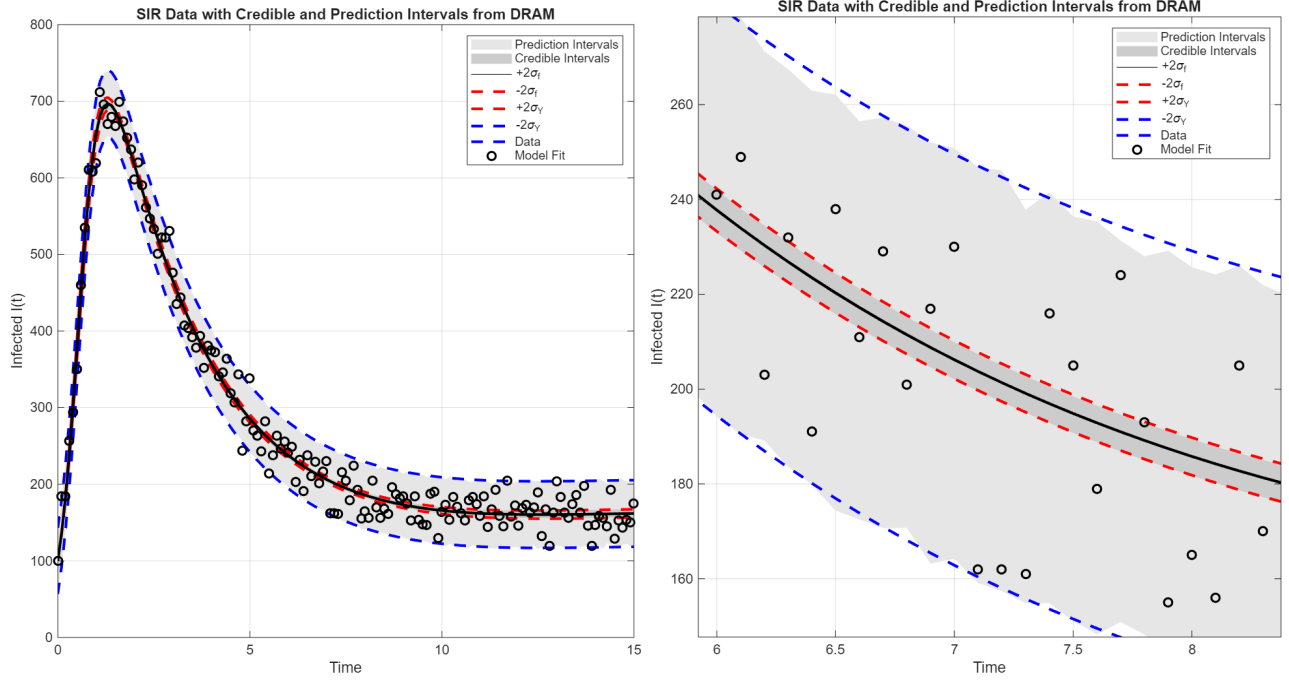


Figure 7: Bayesian Credible and Prediction Intervals for SIR Model from Influenza Data

Figure 8: Bayesian vs Linearization for SIR Model



Solution

We note our nominal parameter values are the best estimates under OLS. We then employ complex-step approximation to find the sensitivities $\frac{\partial I}{\partial \theta_j}(t_i, \theta^*)$, by shifting a parameter by a small complex step $h = 1 \times 10^{-16}$, propagating through our model, and taking the imaginary part of the resulting $I(t_i)$. We use these to construct our sensitivity matrix $\mathbf{S}$. We then compute $\mathbf{V} = \sigma^2(\mathbf{S}^T \mathbf{S})^{-1}$, and calculate $\text{var}[\mathbf{f}(\theta)] = \mathbf{S} \mathbf{V} \mathbf{S}^T$, $\text{var}[\mathbf{Y}] = \mathbf{S} \mathbf{V} \mathbf{S}^T + \sigma^2 \mathbf{I}_{n \times n}$. We compute $\sigma_{\mathbf{f}}(t_i)$ and $\sigma_{\mathbf{Y}}(t_i)$ as the square roots of the diagonal elements of each of these respective matrices, as stated. Figure 8 shows the linearly approximated 2σ intervals for the model response and observations, graphed on top of the Bayesian credible and prediction intervals from Problem 3a constructed by DRAM. A zoomed portion is also shown. We note that the linearization very closely follows the Bayesian intervals, and appears to have very similar levels of comparison to the data. We can conclude that in this case, the linearization has similar accuracy to the Bayesian approach. That is, we can approximate that the $\sigma_{\mathbf{Y}}(t_i)$ intervals have probability of at least 95% of containing a future observable Y_{x^*} .